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Slotted bolt cleat binder

Abstract

A cleat binder is used to secure cables for short circuit protection. The cleat binder includes a yoke, a slotted hex head bolt, a nut, and a strap. The slotted hex head bolt has a head and a slot extending through the bolt. The yoke includes a bottom with a front, a back, a first side, and a second side. Side walls extend from the first and second side of the bottom of the yoke. The slotted hex head bolt extends through openings in the side walls of the yoke. The nut secures the slotted hex head bolt on the yoke. The strap is fed through the slot of the slotted hex head bolt, wrapped around cables, and tensioned to secure the cleat binder to the cables.

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Background/Summary

FIELD OF THE INVENTION
(1) The present invention relates to strap cleat for short circuit protection, and more particularly to a slotted bolt cleat binder for short circuit protection.
BACKGROUND OF THE INVENTION
(2) Conventional cable cleats have been typically used to secure wire or cable bundles to panels, ladder racks, or similar structural support members. The cable cleats include a mounting surface and a body portion that receives the wires or a cable bundle. It is also known to use MS 75 strap cleats to secure cables for short circuit protection. While the known cleats provide desirable characteristics for certain applications, they still have drawbacks and are capable of improvement. Conventional cable cleats are generally difficult and time consuming to install.
(3) It is desirable to provide an improved strap cleat solution for short circuit protection that is lower in cost and easier to install. It is also desirable to provide a strap cleat solution that is flexible and accommodates a range of cable sizes.
SUMMARY OF THE INVENTION
(4) The present invention is directed towards a cleat binder to secure cables and protect cables during short circuit events. The cleat binder includes a yoke, a slotted hex head bolt, a nut, and a strap. The slotted hex head bolt has a head and a slot extending through the bolt. The yoke includes a bottom with a front, a back, a first side, and a second side. A first side wall extends from the first side of the bottom and a second side wall extends from the second side of the bottom. The slotted hex head bolt extends through the yoke and the nut secures the slotted hex head bolt on the yoke. The strap is fed through the slot of the slotted hex head bolt and wrapped around the cables to secure the cleat binder to the cables.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1**A is a perspective view of one embodiment of a slotted bolt cleat binder installed on a trefoil cable arrangement.
(2) FIG. **1**B is a front view of the slotted bolt cleat binder of FIG. **1**A.
(3) FIG. **2** is a partially exploded view of the slotted bolt cleat binder of FIG. **1**A prior to installation.
(4) FIG. **3** is a perspective view of an assembled slotted bolt cleat binder of FIG. **2**.
(5) FIG. **4**A is a front view of the assembled slotted bolt cleat binder of FIG. **2** without the strap.
(6) FIG. **4**B is a front view of the assembled slotted bolt cleat binder of FIG. **4**A with the fastener tightened.
(7) FIG. **5** is a front right perspective view of the slotted bolt cleat binder of FIG. **3** without the strap.
(8) FIG. **6** is a front left perspective view of the slotted bolt cleat binder of FIG. **3** without the strap.
(9) FIG. **7** is a perspective view of the slotted bolt cleat binder of FIG. **5** and a strap being installed around a trefoil cable arrangement.
(10) FIG. **8** is a perspective view of a second embodiment of a slotted bolt cleat binder installed on a trefoil cable arrangement.
(11) FIG. **9** is a partially exploded left perspective view of the slotted bolt cleat binder of FIG. **8**.
(12) FIG. **10** is a partially exploded right perspective view of the slotted bolt cleat binder of FIG. **8**.
(13) FIG. **11** is a perspective view of the slotted bolt cleat binder of FIG. **8**.
(14) FIG. **12** is a perspective view of the slotted bolt cleat binder of FIG. **11** with the strap feed through the slotted bolt cleat binder.
(15) FIG. **13** is a perspective view of a third embodiment of a slotted bolt cleat binder installed on a trefoil cable arrangement.
(16) FIG. **14** is a partially exploded left perspective view of the slotted bolt cleat binder of FIG. **13**.
(17) FIG. **15** is a partially exploded right perspective view of the slotted bolt cleat binder of FIG. **13**.
(18) FIG. **16** is a perspective view of the slotted bolt cleat binder of FIG. **13**.

(19) FIG. **17** is a perspective view of the slotted bolt cleat binder of FIG. **16** with the strap feed through the slotted bolt cleat binder.

DETAILED DESCRIPTION

(20) FIGS. **1**A-**7** illustrate a first embodiment of the slotted bolt cleat binder **30** of the present invention. FIGS. **1**A-**1**B illustrate the slotted bolt cleat binder **30** secured around a trefoil cable arrangement **100**. As illustrated in FIG. **2**, the slotted cleat binder **30** includes a yoke **32**, a slotted hex head bolt **60**, a hex nut **80**, and a strap **90**. The yoke **32** includes a bottom **34** having a front **36**, a back **38**, a first side **40** and a second side **42**. A first side wall **44** extends from the first side **40** perpendicular to the bottom **34** and the second side wall **52** extends from the second side **42** perpendicular to the bottom **34**. The side walls **44**, **52** include inner walls **46**, **54** with a cross knurling pattern and outer walls **48**, **56**, respectively. The first side wall **44** and the second side wall **52** each include an opening **50**, **58** therethrough. The openings **50**, **58** are aligned and are designed to receive the slotted hex head bolt **60**.

(21) The slotted hex head bolt **60** includes a hex head **62** at one end, a slot **64** through the bolt that extends half the length of the bolt **60**, and threads **66** at the end opposite the head **62**.

(22) As illustrated in FIG. **3**, once the slotted hex head bolt **60** has been installed through the openings **50**, **58** in the side walls **44**, **52**, the hex nut **80** is threaded on the bolt **60**. FIGS. **4**A and **4**B illustrate the slotted hex head bolt **60** installed in the yoke **32** without a strap **90**. As the nut **80** is tightening, the side walls **44**, **52** of the yoke **32** bend towards a center of the yoke **32** (see FIG. **4**B). FIGS. **5** and **6** illustrate the inner wall **46**, **54** of each side wall **44**, **52**, respectively, with the knurled surface.

(23) To install the slotted bolt cleat binder **30**, the cleat binder assembly **30** is positioned on a trefoil cable arrangement **100** or a cable bundle. The cleat binder **30** may also be positioned on a cushion sleeve (not illustrated) that has been wrapped around the cable or cable bundle. The strap **90** is wrapped around the trefoil cable arrangement **100** and threaded through the slot **64** in the slotted hex head bolt **60**, as shown in FIG. **7**. Using common tools, the slotted hex head bolt **60** is rotated to wrap the strap **90** around the slotted hex head bolt **60** to tension the strap **90**.

(24) Once the desired tension has been reached, the slotted hex head bolt **60** is held in a tensioned state and the nut **80** is tightened to complete the installation. The action of tightening the nut **80** locks the cleat binder **30** in a number of ways. The slotted hex head bolt **60** is held in place after tightening the nut **80** due to the friction between the hex head **62** of the bolt **60**, nut **80**, and outer walls **48**, **56** of the side walls **44**, **52** of the yoke **32**. The tightening of the nut **80** also bends the side walls **44**, **52** inward applying pressure to the edges of the strap **90** (see FIG. **1**B). The inner walls **46**, **54** of the side walls **44**, **52** of the yoke **32** are knurled to further lock the strap **90** to the yoke **32** and prevent any movement during a short circuit event. The knurled inner walls **46**, **54** lock the strap **90** by concentrating the force provided by the bolt **60** and nut **80** and deforming the strap material. The knurling on the inner walls **46**, **54** may be a cross pattern or a straight pattern.

(25) The slotted bolt cleat binder **30** includes a three-way locking system with the bolt **60** and nut **80**, the strap **90** to the yoke **32**, and the strap **90** to the knurls on the inside walls **46**, **54**. The slotted bolt cleat binder **30** is easily installed by common tools with gloves even in confined spaces. The slotted bolt cleat binder **30** receives various sized cables or cable bundles and is inexpensive to manufacture.

(26) FIGS. **8-12** illustrate a second embodiment of a slotted bolt cleat binder **130**. FIG. **8** illustrates the slotted bolt cleat binder **130** installed on a trefoil cable arrangement **200**. As illustrated in FIGS. **9** and **10**, the slotted cleat binder **130** includes a yoke **132**, a slotted hex head bolt **160**, axial ratcheting teeth **168**, **172**, a tensioner spring **176**, a hex nut **180**, and a strap **190**. The yoke **132** includes a bottom **134** having a front **136**, a back **138**, a first side **140**, and a second side **142**. A first side wall **144** extends from the first side **140** perpendicular to the bottom **134** and the second side wall **152** extends from the second side **142** perpendicular to the bottom **134**. The first side wall **144** and the second side wall **152** each include an inner wall **146**, **154**, an outer wall **148**, **156**, and an opening **150**, **158** therethrough, respectively. The openings **150**, **158** are aligned and are designed to receive the slotted hex head bolt **160**.

(27) The slotted hex head bolt **160** includes a hex head **162** at one end, a slot **164** through the bolt that extends half the length of the bolt **160**, and threads **166** at the end opposite the head **162**.
(28) Axial ratcheting teeth **168**, **172** are welded to the bolt head **162** and the outer wall **148** of the first side wall **144** of the yoke **132**, respectively. The ratcheting teeth **168**, **172** include angled ramps **170**, **174** that become engaged or disengaged through rotation of the hex head **162**. The tensioner spring **176** is positioned adjacent to the outer wall **156** of the second side wall **152** of the yoke **132**.
(29) To install the cleat binder **130**, the cleat binder **130** is positioned on a cushion sleeve (not illustrated) wrapped around a trefoil cable arrangement or cable bundle or the cleat binder **130** is positioned on the trefoil cable arrangement **200** or other cable bundle. The strap **190** is threaded through the slot **164** in the slotted hex head bolt **160**. Using common tools, the slotted hex head bolt **160** is then rotated to wrap the strap **190** around the slotted hex head bolt **160** to tension the strap **190**. Rotating the bolt **160** causes the ratcheting teeth **168**, **172** to disengage by sliding on the angled ramp face **170**, **174** until rotated to the next index position where the teeth re-engage and lock in place. Once the desired tension has been reached and the installation is complete, the spring **176** will apply force ensuring the ratcheting teeth **168**, **172** will remain in an engaged position.
(30) The slotted bolt cleat binder **130** includes a one step process for tensioning and locking a strap around a cable or cable bundle. The slotted bolt cleat binder **130** is easily installed by common tools with gloves even in confined spaces. The slotted bolt cleat binder **130** receives various sized cables or cable bundles and is inexpensive to manufacture.
(31) FIGS. **13-17** illustrate a third embodiment of a slotted bolt cleat binder **230**. FIG. **13** illustrates the slotted bolt cleat binder **230** installed on a trefoil cable arrangement **300**. As illustrated in FIGS. **14** and **15**, the slotted cleat binder **230** includes a yoke **232**, a slotted hex head bolt **270**, a compression spring **278** or a conical disc spring (not illustrated), a hex nut **280**, and a strap **290**. The yoke **232** includes a bottom **234** having a front **236**, a back **238**, a first side **240**, and a second side **242**. A first side wall **244** extends from the first side **240** perpendicular to the bottom **234** and the second side wall **252** extends from the second side **242** perpendicular to the bottom **234**. The first side wall **244** and the second side wall **252** each include an inner wall **246**, **254**, an outer wall **248**, **256**, and an opening **250**, **258** therethrough, respectively. The openings **250**, **258** are aligned and are designed to receive the slotted hex head bolt **270**.
(32) The slotted hex head bolt **270** includes a hex head **272** at one end, a slot **274** through the bolt **270** that extends half the length of the bolt **270**, and threads **276** at the end opposite the head **272**.
(33) The first side wall **244** includes narrowing indentations **260** and rectangular locators **262** extending from the outer side wall **248** away from the yoke **232**. The rectangular locators **262** frame the upper portion of the first side wall **244** to capture the hex head **272** of the slotted hex head bolt **270**.
(34) To install the cleat binder **230**, the cleat binder **230** is positioned on a cushion sleeve (not illustrated) wrapped around a trefoil cable arrangement or other cable bundle or the cleat binder **230** is positioned on the trefoil cable arrangement **300** or other cable bundle. The strap **290** is threaded through the slot **274** in the slotted hex head bolt **270**. As illustrated in FIG. **16**, the spring **278** is sufficiently powerful to stop the hex head **272** of the bolt **270** from prematurely engaging the rectangular locators **262**. The strap **290** can be tightened around the trefoil cable arrangement **300** or other cable bundles using a common tool, such as a standard wrench. Once sufficiently tight, a second wrench is used to tighten the nut **280**. This action compresses the spring **278** and forces the hex head **272** of the bolt **270** to be captured between the rectangular locators **262**. The illustrated design shows three rectangular locators **262** spaced 60 degrees from one another to match the facets of the hex head **272** of the slotted hex head bolt **270**, but the design could have any number of locators from 1 to 6 to interact with the hex head **272** of the slotted hex head bolt **270**.
(35) Once the slotted hex head bolt **270** has been tightened, the hex head **272** is retained and will resist releasing. The slotted hex head bolt **270** could be reopened to allow for more cables to be installed and then retightened to secure the slotted bolt cleat binder **230**, when desired. The slotted bolt cleat binder **230** is easily installed by common tools with gloves even in confined spaces. The

slotted bolt cleat binder **230** receives various sized cables or cable bundles and is inexpensive to manufacture.
(36) Furthermore, while the particular preferred embodiments of the present invention have been shown and described, it will be obvious to those skilled in the art that changes and modifications may be made without departing from the teaching of the invention. The matter set forth in the foregoing description and accompanying drawings is offered by way of illustration only and not as limitation. The actual scope of the invention is intended to be defined in the following claims when viewed in their proper perspective based on the prior art.

Claims

1. A cleat binder for securing cables, the cleat binder comprising: a yoke including a bottom with a front, a back, a first side, and a second side; a first side wall extends from the first side of the bottom, and a second side wall extends from the second side of the bottom, the first side wall and the second side wall each include an outer wall and an inner wall, wherein the inner wall of the first side wall and the inner wall of the second side wall are knurled; a slotted hex head bolt extending through the yoke, the slotted hex head bolt having a head and a slot extending through the bolt; a nut for securing the slotted hex head bolt on the yoke; and a strap fed through the slot of the slotted hex head bolt for securing the cleat binder to cables.
2. The cleat binder of claim 1, wherein the first side wall and the second side wall each include an opening therethrough.
3. The cleat binder of claim 2, wherein the opening in the first side wall and the opening in the second side wall are aligned, and wherein the slotted hex head bolt extends through the opening in the first side wall and the opening in the second side wall.
4. The cleat binder of claim 1, wherein as the nut is tightened on the slotted hex head bolt, the first side wall of the yoke and the second side wall of the yoke bend towards a center of the yoke.
5. The cleat binder of claim 1, further comprising axial ratcheting teeth and a spring.
6. The cleat binder of claim 5, wherein the axial ratcheting teeth are affixed to the head of the slotted hex head bolt and an outer wall of the first side wall of the yoke.
7. The cleat binder of claim 6, wherein the axial ratcheting teeth include angled ramps that become engaged or disengaged through rotation of the slotted hex head bolt.
8. The cleat binder of claim 5, wherein the spring is positioned adjacent to an outer wall of second side wall of the yoke.
9. The cleat binder of claim 5, wherein the axial ratcheting teeth are affixed to the head of the slotted hex head bolt and an outer wall of the first side wall of the yoke, and wherein the spring is positioned adjacent to an outer wall of the second side wall, whereby the spring applies a force to maintain the axial ratcheting teeth in an engaged position.
10. The cleat binder of claim 1, further comprising a compression spring positioned on the slotted hex head bolt.
11. The cleat binder of claim 1, wherein the first side wall has narrowing indentations and rectangular locators, the rectangular locators frame an upper portion of the first side wall.
12. The cleat binder of claim 11, wherein the rectangular locators capture the head of the slotted hex head bolt.
13. The cleat binder of claim 1, wherein a compression spring is positioned on the slotted hex head bolt adjacent to the head and an outer wall of the first side wall of the yoke, and wherein the outer wall of the first side wall of the yoke has rectangular locators extending therefrom, whereby once the strap has been tightened around cables, the nut is tightened on the slotted hex head bolt to compress the spring and force the head of the slotted hex head bolt to be captured by the rectangular locators.
14. A cleat binder for securing cables, the cleat binder comprising: a yoke including a bottom with a front, a back, a first side, and a second side; a first side wall extends from the first side of the bottom, and a second side wall extends from the second side of the bottom; a slotted hex head bolt extending

through the yoke, the slotted hex head bolt having a head and a slot extending through the bolt; a nut for securing the slotted hex head bolt on the yoke; a strap fed through the slot of the slotted hex head bolt for securing the cleat binder to cables; wherein as the nut is tightened on the slotted hex head bolt, the first side wall of the yoke and the second side wall of the yoke bend towards a center of the yoke; and wherein the first side wall has an inner wall with a knurled surface and the second side wall has an inner wall with a knurled surface, whereby the knurled inner walls engage the strap to prevent movement during a short circuit event.